AQUASHIELD CPS

Physics-Informed Cyber-Physical Attack Localization & Resilient Recovery

Sustainathon 2024 • VIT Vellore
UN SDG 6 & SDG 9 • Hackathon Presentation Dossier

> **Official Problem Statement:**
> *"The Attack-Localization Problem — Detecting an anomaly does not necessarily reveal whether its origin is a sensor, communication channel, controller, firmware component or physical actuator. Cybersecurity systems can identify abnormal behaviour without providing sufficient information about where an attack originated or how it propagated through a cyber-physical system."*

1. Executive Summary in Layman's Terms

Think of modern sustainable infrastructure (like smart water treatment facilities, desalination plants, or solar microgrids) like the human body: sensors are the eyes, communication wires are the nerves, the controller (PLC) is the brain, firmware is the brain's instincts, and actuators (pumps/valves) are the muscles.

The Problem: When a cybersecurity alarm goes off, existing tools only know *"something feels wrong"*. They cannot tell if a thermometer is lying, a nerve wire was intercepted, the brain was hacked, or a muscle is jammed. In a water plant, guessing wrong causes panic operators to shut down the plant and dump hundreds of thousands of liters of clean drinking water into the sewer.

The AquaShield Solution: Hackers can forge digital network packets or fake sensor readings, but **they cannot violate the laws of physics**. AquaShield uses a real-time Physics Digital Twin to cross-verify mass conservation equations ($\Delta V = \int (Q_{in} - Q_{out}) dt$). In under **38 milliseconds**, it isolates the exact guilty layer, quarantines it, and engages a virtual estimator so the plant never stops producing clean water.

2. Live Running Application: Visual Dashboard



Figure 1: AquaShield CPS running live — L1 Sensor Spoofing attack injected, causal red laser locking onto Sensor SEN-204, and the 5-layer localization matrix isolating the breach in 38ms while the plant remains 99.4% operational.

3. The 5-Layer CPS Architecture (Direct Prompt Mapping)

Our system explicitly monitors, models, and isolates each of the five layers highlighted in the hackathon prompt:

CPS Layer	Facility Component	Attack Scenario Implemented	AquaShield Localization Proof
1. Sensor	SEN-204 (Level Probe on TNK-102)	**Telemetry Spoofing:** Reading frozen at 32% while raw inflow is filling tank toward overflow.	Mass Conservation Invariant: Observed $dh/dt = 0$, but flow meter proves $Q_{in} - Q_{out} = +3.8$ L/s. Physics violation isolated to sensor.
2. Channel	COMM-BUS-03 (Modbus TCP Port 502)	**Replay & Jitter Attack:** Injects 385ms network delay and replays stale telemetry timestamps.	Temporal Consistency Audit: Packet timestamp skew $\Delta t > 250$ms and sequence jumps isolate the network bus.
3. Controller	PLC-CORE-1 (PID Setpoint Block)	**Setpoint Hijack:** Attacker overrides membrane pressure target from 3.2 bar to dangerous 8.5 bar.	Safety Envelope Violation: Target setpoint exceeds physical hydraulic burst limit (5.0 bar) without authorized token.
4. Firmware	FW-ROM-0x7C (Turbine Governor Microcode)	**Stuxnet Rootkit:** Corrupt microcode hides pump oscillations to cause motor cavitation failure.	Hardware Root-of-Trust (RoT): SHA-256 cryptographic hash mismatch (Expected 0x8F9A42... received 0xDEADBEEF).
5. Actuator	ACT-VALVE-202 (Emergency Bleed Solenoid)	**Mechanical Seizure:** Controller signaled 0% (CLOSED), but valve is physically jammed open at 100%.	Actuator Discrepancy Invariant: Command voltage = 0V, but downstream discharge flow velocity = 65.0 L/s.

4. Core Dashboard Features Explained

A. Interactive SCADA Digital Twin

Simulates a 4-tank continuous water recycling facility:

- **TNK-101 (Raw Intake):** Inflow sedimentation basin.
- **TNK-102 (Filtration):** Membrane filtration (Houses SEN-204).
- **TNK-103 (RO & Disinfection):** Chlorine dosing & reverse osmosis.
- **TNK-104 (Potable Reservoir):** Clean drinking water supply.

Dynamic Feature: Live animated SVG liquid levels, rotating pump turbines, and red laser tracing to the compromised node.

B. Traditional IDS vs. AquaShield Comparison

Side-by-side benchmark showing the sustainability advantage:

Traditional IDS: Blind alert $\to$ emergency plant shutdown $\to$ 150,000L clean water dumped $\to$ 4-8 hours manual technician inspection.

AquaShield: Localized in 38ms $\to$ quarantines only the bad node $\to$ switches to virtual estimator $\to$ 0 seconds downtime $\to$ 0L water dumped.

C. Chaos Injection Studio (Judge Testing Dock)

5 dedicated one-click buttons to inject any of the 5 layer attacks live in front of judges, plus a **Restore Nominal Baseline** button that resets the system with sound effects and screen confetti.

D. Live SCADA Audit & Math Inspector

Real-time industrial event log stream with precise timestamps, alongside dynamic evaluation of the mass balance conservation formula: `ΔV = ∫ (Q_in − Q_out) dt = A × (dh/dt)`

AQUASHIELD CPS

5. How to Operate the Demo Tomorrow (Step-by-Step)

Preparation Checklist (5 Minutes Before Judges Arrive)

1. Open Terminal on your laptop and run: `cd /Users/pradyun/Desktop/SUSTAINATHON && npm run dev`
2. Open Google Chrome to: **http://localhost:5173**
3. Press **F11** (or `Cmd + Shift + F` on Mac) to enter full-screen mode for the clean SCADA look.
4. Set laptop volume to ~40% so cyber beeps, alarm sirens, and success chimes are clearly audible.
5. Ensure the screen is in the **Nominal Baseline** state (all indicators green).

6. The 2-Minute Winning Pitch Script (Rehearse This!)

STEP 1: The Hook & The Problem — 0:00 - 0:30

"Good morning judges! We are presenting AquaShield CPS, solving the Attack-Localization Problem in critical sustainable infrastructure.
In smart water plants and solar grids, standard cybersecurity systems can detect that something is wrong, but they cannot tell you WHERE the attack started — whether a sensor is lying, a network packet was intercepted, a PLC controller was hacked, firmware was modified, or a physical valve is jammed.
Because operators cannot localize the root cause, they panic, shut down the facility, and dump hundreds of thousands of liters of clean drinking water."

STEP 2: The Interactive Test (Hand the Mouse to the Judge!) — 0:30 - 0:45

"To show you how we solve this, judge, please click any button in our Chaos Injection Studio below — for example, 'Layer 1: Sensor Spoofing'."

STEP 3: The Localization Proof (The 'Aha!' Moment) — 0:45 - 1:15

"Watch what just happened:
1. An alarm sounded, and our red causal laser locked onto Sensor SEN-204.
2. In our 5-Layer Stack on the right, ONLY Layer 1 is isolated. The other 4 layers are proven completely safe.
3. How did we know? Physics! Inflow minus Outflow mathematically requires the tank level to be at 73%, but the hacker locked the sensor reading at 32%.
4. In 38 milliseconds, AquaShield quarantined the sensor and engaged our Digital Twin Virtual Estimator."

STEP 4: The Sustainability Punchline & Finale — 1:15 - 2:00

"A traditional IDS would have dumped 150,000 Liters of water and caused 6 hours of downtime. With AquaShield, plant availability stays at 99.4%, clean drinking water keeps flowing, and zero water is wasted.
(Click 'Restore Nominal Baseline' -> victory chime plays & confetti triggers!)
And with one click, our physical invariants restore nominal operations. Thank you!"

7. Answers to the Top 3 Judge Questions

Q1: How is this different from a normal Intrusion Detection System (IDS)?
Answer: "Standard IDS only looks at cyber data (CPU, network traffic). They treat physical pumps and water tanks as a black box. AquaShield combines cyber data with physical invariants (mass balance & fluid laws) to tell the difference between a sensor lying vs. an actuator physically breaking."

Q2: Why is it demoed on software instead of hardware?
Answer: "Industrial water plants handle tens of thousands of liters of pressurized water. Our software is an industrial-grade Digital Twin running exact ordinary differential equations (ODEs) of fluid mechanics, which is the exact industry-standard architecture used by Siemens and GE Digital."

Q3: What if the attacker targets the network channel instead of the sensor?
Answer: "When Layer 2 (Network) is attacked, the sensor reading is physically plausible, but packet latency spikes to 385ms and sequence timestamps jump. AquaShield isolates Layer 2 and reroutes traffic to a redundant industrial mesh channel in milliseconds."